nature cities

Article

https://doi.org/10.1038/s44284-024-00119-4

A universal framework for inclusive 15-minute cities

Received: 6 February 2024

Accepted: 2 August 2024

Published online: 16 September 2024

Check for updates

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Proximity-based cities have attracted much attention in recent years. The '15-minute city', in particular, heralded a new vision for cities where essential services must be easily accessible. Despite its undoubted merit in stimulating discussion on new organization of cities, the 15-minute city cannot be applicable everywhere, and its very definition raises a few concerns. Here we tackle the feasibility and practicability of the 15-minute city model in many cities worldwide. We provide a worldwide quantification of how close cities are to the ideal of the 15-minute city. To this end, we measure the accessibility times to resources and services, and we reveal strong heterogeneity of accessibility within and across cities, with a pivotal role played by local population densities. We provide an online platform (https://whatif.sonycsl.it/15mincity/) to access and visualize accessibility scores for virtually all cities worldwide. The heterogeneity of accessibility within cities is one of the sources of inequality. We thus simulate how much a better redistribution of resources and services could heal inequity by keeping the same resources and services or by allowing for virtually infinite resources. We highlight pronounced discrepancies among cities in the minimum number of additional services needed to comply with the 15-minute city concept. We conclude that the proximity-based paradigm must be generalized to work on a wide range of local population densities. Finally, socio-economic and cultural factors should be included to shift from time-based to value-based cities.

In the past few years, a strong push for redesigning our urban landscapes emerged to set the stage for a future of more sustainable and connected urban living. This novel movement took off from the need for local accessibility to services and reducing carbon emissions from private transportation.

The recently trending concept of the 15-minute city contends that cities can function more effectively, equitably and environmentally if essential services and key amenities are within 15 minutes by non-polluting forms of transport, such as walking or cycling[1]. It provides an alternative to car-dependent urban designs and generally deleterious long-distance commuting. By focusing on proximity, this approach can address pressing issues like air pollution, traffic congestion and social disparities[2]. The 15-minute city concept also rose in popularity during and after the COVID-19 pandemic, as local communities regained importance and people, even reluctantly, rediscovered the possibility of enjoying their neighborhood[3], calling for new urban solutions to enhance walkability[4].

The notion of proximity-based cities has been around for a while, due to decades of increasing interest in compact cities[5,6]. The long-standing dualism between suburban cities and compact urban planning

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https://doi.org/10.1038/s44284-024-00119-4

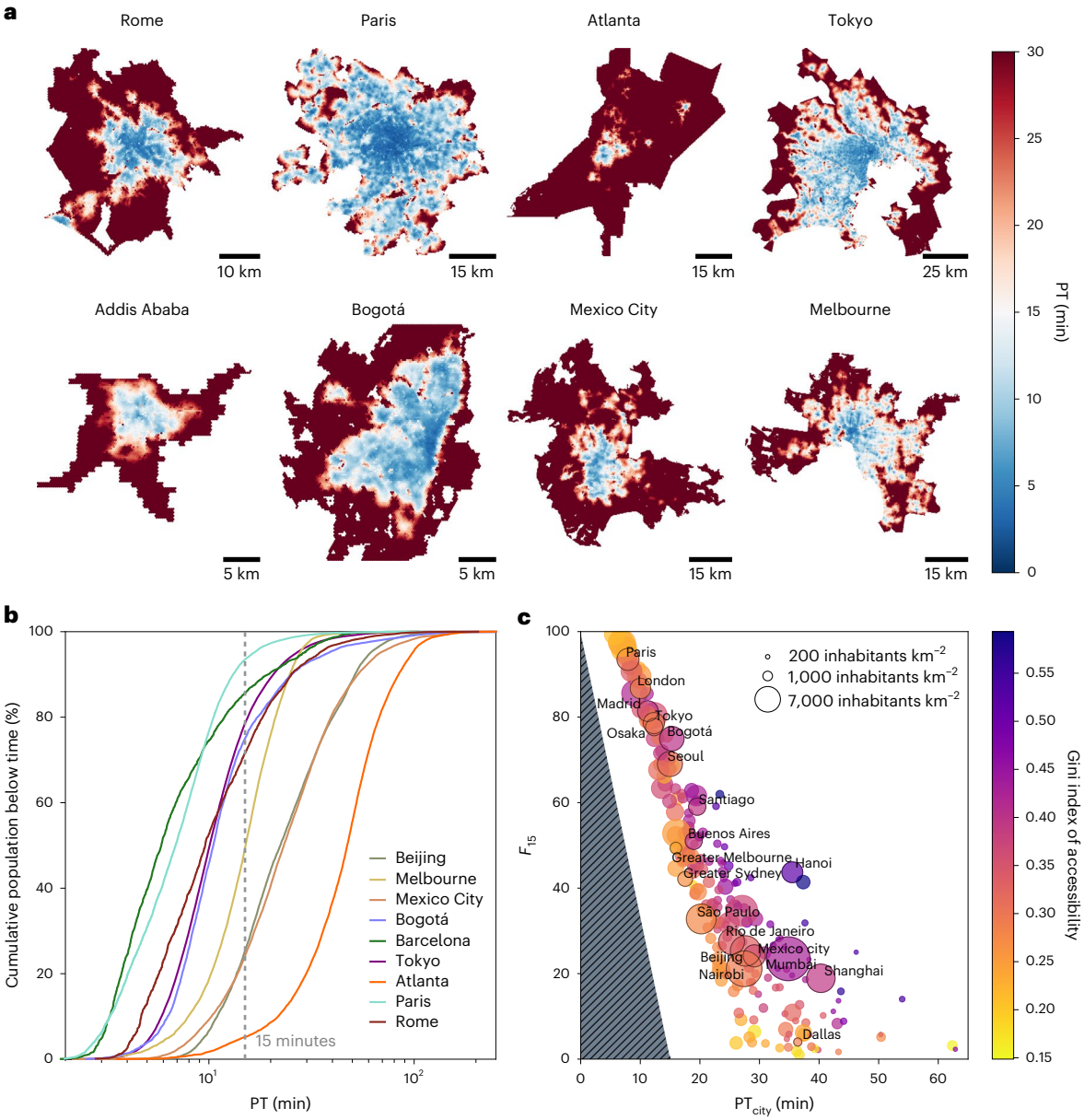


Fig. 1 | Local accessibility measured in several different cities. a, Computed maps of local accessibility scores for a subset of the studied cities. Each hexagon in each city is colored according to PT, that is, the time to access the services of the 15-minute city. The color code is such that blue (red) colors correspond to areas where the accessibility time of the services is below (above) 15 minutes. Remarkable differences are observed among different cities. For instance, a small fraction of the city area in Atlanta has PTs that are less than 15 minutes. Conversely, Paris has a considerable fraction of the city area below 15 minutes. **b**, The cumulative distribution of PT scores among the population in selected cities, that is, the population fraction whose accessibility is below a given PT. The curves for different cities display similar S-shaped behaviors, though their growth occurs at very different positions. We highlight the curve intercept for a PT of 15 minutes, revealing the fraction of the population living in a 15-minute condition. **c**, A scatter plot of different accessibility scores of cities in our study. For each city, we report the average PT score, PT_{city}, versus the fraction of the population living in a 15-minute condition, F_{15}. The circle size is proportional to the population density of the city (inhabitants per km^2). The color codes for the Gini index, which captures the unequal accessibility across cities. The triangular gray area represents a theoretically unreachable phase: if a city has a low PT_{city}, the proportion of people within 15-minute accessibility cannot be lower than a certain percentage. We observe a decreasing trend of F_{15} versus PT_{city}, which is to be expected. Less trivial is that the Gini index shows an increasing trend with PT_{city}.

has been at the center of public debate recently due to the need to create sustainable environments and reduce CO_2 produced by transport emissions, which recent studies confirm to be lower in dense cities[7]. Therefore, a new focus on eco-urbanism solutions has arisen, with the challenge of building more compact cities and improving accessibility[8].

Transforming a city into a 15-minute one is a challenging task. It calls for substantial shifts in urban design, transportation strategies and land use policies[9]. Despite its increasing appeal, the 15-minuteness raises essential concerns. For instance, a perfectly 15-minute city could be highly unequal regarding the quality of the services provided or might lead to a reduction of green areas[10]. Two areas of the same city could have perfect 15-minuteness, but one could have top-tier and the other inferior services, forming a fertile ground for inequality and segregation. In addition, as we shall see in this Article, local population density plays a pivotal role in determining the best scenarios for future cities. The practicality of the 15-minute city still needs to be determined, as well as its consequential impact on the resident's quality of life. Some studies have begun to quantitatively measure the impact and influence

of such an urban structure[11]. Moreover, many cities have begun journeys to improve accessibility and create compact districts[12], but there is no unique recipe for such changes.

Measuring local accessibility to services is a cornerstone in this exploration. The quest for measuring accessibility began long ago, and various measures have been employed[13–15]. All these studies point out that cities are not homogeneous bodies: it is crucial to acknowledge that not all parts of a city are equal, leading to varying degrees of accessibility[16–20]. Thus, previous studies point towards inequality of access even in 15-minute cities[21,22]. Therefore, urban studies have focused on population landscapes, that is, the distributions of local population density and how to improve accessibility for the largest possible fraction of the population. One popular approach is to place amenities to meet the needs of the population optimally; this complex problem can be formulated in various ways, and some algorithms have been proposed. For instance, in ref. 23 the total travel distance of the population to reach the nearest facility in space is minimized, simulating this allocation at a country level, and in a similar spirit in ref. 24 the optimization of the facilities is performed at the scale of a single city. Other previous work includes the use of shape grammar[25] and considers mobility[26]. Here, we aim to merge these theoretical accessibility designs with the quest for the 15-minute city to understand how optimally placed amenities can improve local accessibility in urban contexts.

As a first contribution of the Article, we measure accessibility in cities worldwide, probing the distribution of services and pinpointing regions where disparities are the most conspicuous. Further, we scrutinize the practicality of the 15-minute city by imagining a more equal distribution of services, considering the spatial distribution of population densities. We propose a novel algorithmic approach to create scenarios wherein city services are redistributed based on population distributions. By simulating a hypothetical equal relocation of services, we can forecast potential enhancements in urban accessibility. Finally, we employ our algorithm to fine-tune the number of services to reach an equal 15-minute city condition. Measuring the quantity of per capita services needed for this goal, we demonstrate that the number of services required fluctuates remarkably and depends on the population density and how the population is distributed in the territory. We contend that our analysis can offer insights that can serve as catalysts for urban planners and decision-makers, fostering the creation of cities that are more equitable, accessible and sustainable.

Results

Assessing current 15-minuteness

We start our analysis by conducting a thorough assessment of the closeness of current cities to the ideal of 15-minuteness. The first notable result, extracted from the study of many cities worldwide, is a pretty substantial heterogeneity of accessibility both within and across different cities. In this Article, we will illustrate more than 50 cities in depth, and we refer the reader to the online platform that we made freely available, which now covers around 10,000 cities worldwide. For all cities, we define and measure the so-called proximity time (PT; see 'Accessibility calculation' for its definition) both locally and for each city as a whole. PT measures how long it takes from a specific point in the city to reach the services of the 15-minute city, either by walking or cycling. Figure 1a reports the maps of PT values (by walking) for some selected cities. The first striking evidence refers to the differences among different cities. Equally, remarkable differences are also observed within a single city, implying that accessibility is not a 'currency' equally distributed in the population, with a centric structure of cities or polycentric one in some notable cases, for example, Barcelona and Paris. The distribution of all PT scores of cities worldwide can be found in Supplementary Section 1.

To quantify the level of inequality in accessibility, we measure, for each city, the fraction of the population living in a 15-minute condition, that is, the fraction of residents with access to essential services within a 15-minute radius. We name this quantity F_{15}. Here essential services are classified in nine categories: outdoor activities, learning, supplies, eating, moving, cultural activities, physical exercise, services and health care. This categorization captures most essential daily activities and allows for both general and category-specific accessibility assessment (see 'Accessibility calculation'). Figure 1b,c shows a quantification of the inequalities in accessibility within and across cities. Two striking results emerge. First, there is considerable variation in the fraction of the population living in a 15-minute condition across different cities (Fig. 1b). Second (Fig. 1c), within the same city, variations in accessibility can be large, and they tend to grow with average PT score, meaning that cities with bad average accessibility are also more unequal. Finally, substantial variations exist among cities in different cultural contexts, mainly based on how car-centric and suburban the urban planning style is.

Our analysis consistently highlights patterns where city centers have better access to services than peripheral areas. However, notable exceptions exist, such as Paris or Barcelona, whose recent policies on increasing local access to services are well known[12]. These cities exhibit a more evenly distributed accessibility, transcending the typical center–periphery divide. It is in the nature of cities, however, to provide more services in their centers: it is the most convenient place for people from different areas to meet, and therefore an attractive place for offices and shops. But the self-reinforcing process of 'rich get richer' has to be avoided: more services mean more people willing or needing to visit the city centers and an ever-increasing demand for services in the centers, making the peripheries ever more empty and isolated.

Towards an optimally equal distribution of opportunities

Having assessed the levels of inequality in urban accessibility, it is natural to raise the question of what could be done to mitigate those inequalities. Lacking systematic empirical examples, we could first ask whether a suitable relocation of the existing points of interest (POIs) could lead to better accessibility patterns and, on a quantitative ground, to more of the population living in a 15-minute condition (higher F_{15} values). To this end, we have devised an algorithm that optimally redistributes the POIs that already exist in a given city. Figure 2a describes the algorithm's rationale for redistributing POIs based on the population distribution within each city. The algorithm aims to have an equal amount of services per capita, or equivalently per 1,000 people, across the city, leading to a more balanced distribution of services than the actual distribution. In this scenario, every amenity will serve an approximately equal amount of people in its neighborhood.

The application of our algorithm yields some insights when considering how many services need to be relocated to different areas from their original ones. Figure 2b illustrates that cities known for their car-centric designs, such as Atlanta and other North American cities, need to relocate a high percentage of POIs, over 70% in some cases; conversely, certain European cities, including Milan, Copenhagen, Lisbon and Paris, already demonstrate a well-optimized, homogeneous distribution of services. Here, the need for relocation is minimal, reinforcing the effectiveness of the present urban design in making the city walkable and accessible. These findings highlight the diverse challenges different cities can face as they strive to improve their accessibility and become 15-minute cities. Figure 2c reports an example of how the relocation algorithm changes the map of the local number of POIs per 1,000 people in Rome.

Let us see now how the relocation of POIs affects accessibility across various cities. We present these shifts in accessibility in Fig. 3 through two distinct visuals. Figure 3a,b shows maps of Rome and Atlanta with a color code associated with the changes in accessibility (the PT score) at the local level. By redistributing POIs, we create areas of positive changes (blueish areas) and negative changes (reddish areas), whereas some areas are unchanged. It is remarkable how the city center does not get stripped of services despite the growth in accessibility of other less central areas. Further, the peripheral regions

https://doi.org/10.1038/s44284-024-00119-4

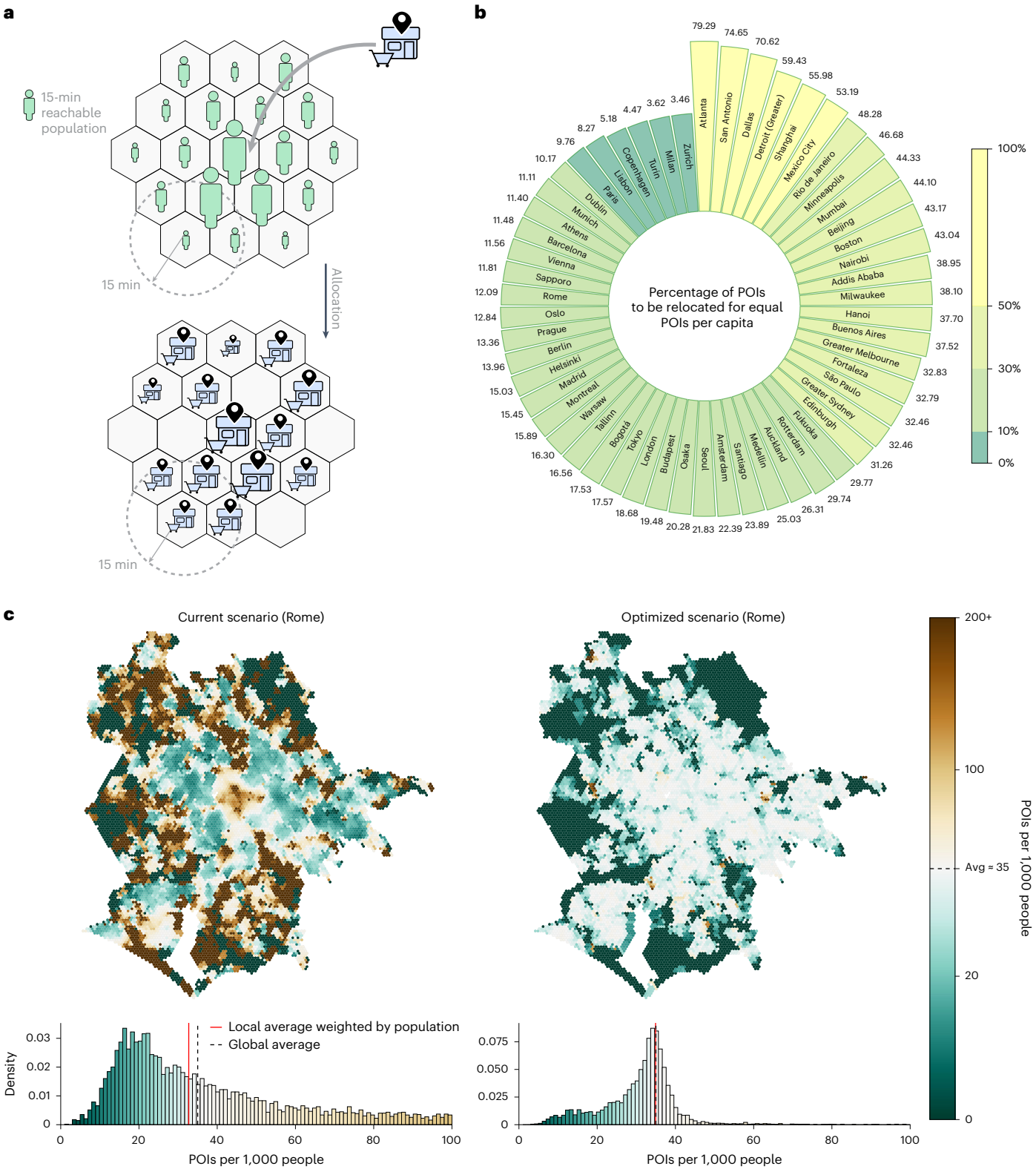


Fig. 2 | Algorithmic distribution of POIs. a, Schematic showing the rationale for the algorithm (see 'POI relocation algorithm' for a thorough description): the more people who are reachable in 15 minutes from a certain area of a city, the more POIs will be placed there and its surroundings. **b**, A ranking of the fraction of POIs relocated in each city after applying the algorithm. A stark variation across cities is immediately noticeable. **c**, Two maps of Rome colored according to the local number of POIs per 1,000 people, equivalent to POIs per capita, calculated before (left) and after (right) application of the relocation algorithm. Each map is complemented by a histogram. The algorithm makes the local availability of services per capita close to the global average in the city.

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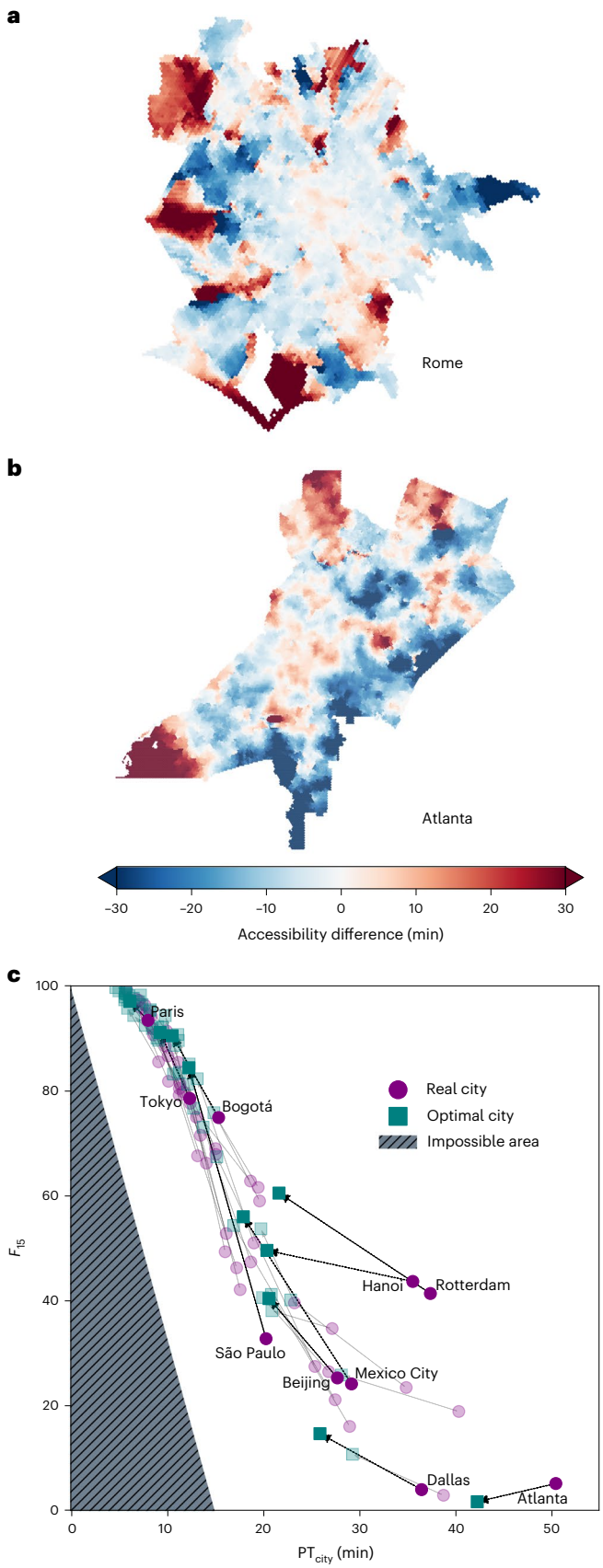


Fig. 3 | Impact of POIs relocation on accessibility. a,**b**, The accessibility difference after applying the algorithm to Rome (**a**) and Atlanta (**b**): red areas get worse because they are over-served or uninhabited. By contrast, blue areas are under-served and consequently improve their accessibility. **c**, The change in the measures of local accessibility depicted in Fig. 1 after applying the algorithm.

experience a substantial enhancement in their access to services, validating the efficiency of the relocation strategy in bridging accessibility disparities. Reddish areas get worse in accessibility, presumably because they were already well served or more likely because they are less densely populated.

Figure 3c reports a scatter plot of F_{15} versus PT_{city}, analogous to that of Fig. 1c. The arrows illustrate the changes a selected set of cities undergo. It is evident how the redistribution of the existing POIs triggers a leftward and upward movement of the cities, that is, a trend towards lower values of PT_{city} (higher average accessibility) and higher F_{15} (larger equality). A notable exception is represented by Atlanta, which warrants a deeper insight. Given the high level of sprawl in Atlanta, the few compact neighborhoods in the center of Atlanta worsen their accessibility. Thus, the city sees a small decline in the proportion of residents within 15-minute areas, despite witnessing an improvement in its average local accessibility. The distributions of accessibility scores before and after the relocation of services can be found in the Supplementary Information.

Our results emphasize the hypothesis that the 15-minute city paradigm is suitable for relatively compact (that is, high-density) urban areas. Even if a strategic relocation of services or an increase of their capillary penetration can notably improve accessibility, the whole paradigm of the 15-minute city has to be rethought in areas with lower urban densities, such as suburban or intermediate zones. We shall return to this point later on.

The quest for inclusive 15-minute cities

So far, we have explored the possibility of better distributing the available opportunities in a city to improve the equality of access to them. Now, we take a step further and ask ourselves about the optimal number and distribution of POIs to make a 15-minute city for the largest population fraction.

We can employ our relocation algorithm to simulate any quantity of POIs in a given city. In particular, we are interested in quantifying the number of POIs required to become a 15-minute city. With this aim in mind, we simulated the optimal city structure considering an increasing number of POIs. We ran our algorithm independently for each POI type (all services are treated equally in our study). This approach allows us to introduce a parameter for gauging a city's accessibility: the number of POIs per 1,000 people.

In Fig. 4, we present the results of our simulations. Figure 4a presents the proportion of the population with a PT below 15 minutes, F_{15}, which features strong variation across cities. We compute, in particular, the intercept of the curves with an F_{15} of 90%, that is, the number of POIs per 1,000 residents necessary for 90% of the city's population to live within a 15-minute access area. Note that this number is not the bare minimum required to have widespread 15-minute accessibility, since the hypothesis of a homogeneous density of POIs per capita is not always satisfied.

The quantity of POIs per capita required for a city to be a '90% 15-minute city' displays considerable variation among the cities studied. With reference to Fig. 4b, for instance, highly dense and homogeneous cities, like Mumbai and Bogotá, require only 0.45 and 0.56 POIs per 1,000 residents, respectively, to achieve the 15-minute city. This occurs similarly in many South American cities such as São Paulo, Mexico City and Fortaleza. Of course, the debate is wide open about the potential overcrowding of POIs and their quality, but this topic falls beyond the scope of our current study.

European and Asian cities present a varied landscape. They lie in an intermediate zone, displaying differing trends based on their urban layout and design. For instance, cities historically designed with a greater focus on automobile transit, like Rotterdam, or with large suburban areas, like Tallinn, Edinburgh, Amsterdam or Helsinki, require a higher number of POIs per capita to become 15-minute cities. This is also the case in cities in Australia and New Zealand. By contrast, compact cities,

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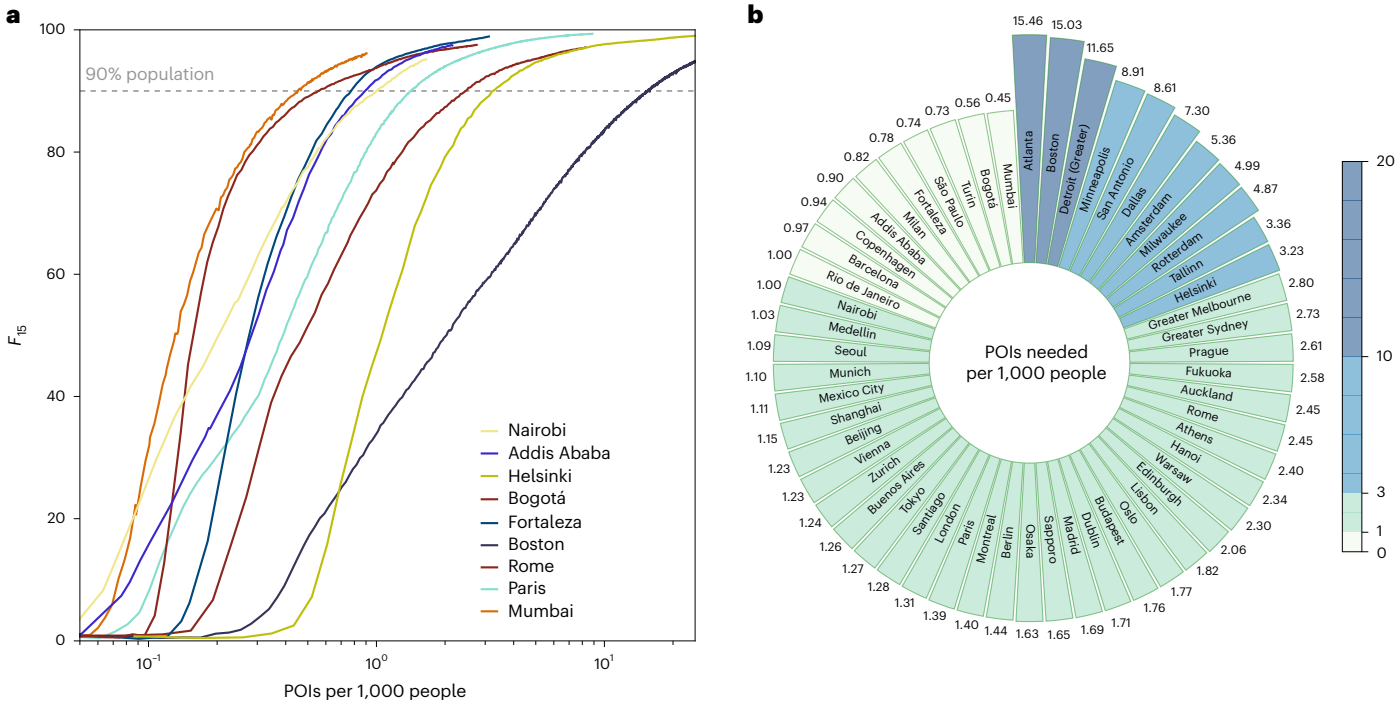


Fig. 4 | Optimal number of POIs per 1,000 people. a, Values of F_{15} as a function of an increasing number of POIs per 1,000 people distributed across the city by our algorithm. The intersection with an F_{15} of 90% defines the number of POIs per 1,000 people needed to become a 15-minute city. **b**, Radial histogram reporting the number of POIs per 1,000 people needed to become a 15-minute city. This number is extremely heterogeneous, resulting from very diverse local and global population densities.

such as Turin, Milan and Barcelona, demand fewer POIs per capita due to their dense and homogeneous urban structure. Another peculiar case is that of cities with discontinuous urban textures, such as Rome or Athens, for which historical heritage and geographical constraints represent an additional challenge towards a compact design.

As already observed, many US cities stand as outliers and represent unique challenges. To achieve 15-minute city status, these cities would require an excessively high number of POIs. Atlanta, for example, would need more than 15 POIs per 1,000 residents, which translates to one POI for every 64 residents. Such a requirement appears unsustainable, once again emphasizing the impact of city density and design on the feasibility of the 15-minute city concept.

Discussion and conclusions

This study contributes to the ongoing conversation on urban planning and design by quantifying local accessibility worldwide and offering an empirical method for simulating scenarios towards proximity-based cities that optimize equal access to opportunities. The contribution of this paper is twofold: methodological and conceptual.

On the methodological side, we have introduced a tool to quantify how close a city is to the 15-minute ideal. We adopted this metric to assess the status of many cities worldwide. Our open-access platform (https://whatif.sonycsl.it/15mincity/) allows everyone to explore cities or portions of them. The second crucial methodological contribution is the conception and implementation of a heuristic algorithm to redistribute the POIs in a city based on equal accessibility to the largest population fraction. Although relocating activities can be a slow process, POIs related to commercial activities represent a very fast-paced environment where many activities can be displaced in a relatively short time, looking for better opportunities. In addition, many cities are promoting policies of reuse and regeneration with frequent changes in the intended use of buildings. We tested the algorithm in two conditions: the relocation of existing POIs and the quest for optimal POIs given the local population densities.

On the conceptual side, this paper provides several essential observations.

First, we provided a global picture of current local accessibility. We drew accessibility maps based on the computation of a new observable, the PT score, which quantifies the time needed from each local area to reach the essential urban services by walking and cycling. Our results highlight a profound heterogeneity in accessibility patterns within and across cities, mirroring different aspects, from geomorphological/ecological to historical/cultural and management. In the comparison among cities, marked variations appear, with only a tiny proportion of the cities very well positioned to become 15-minute cities and an extensive distribution on more or less large values of the score. Stark contrasts are also found in different areas of the same city, with the inequality of accessibility usually following a core-periphery gradient and a few cities that seem to have more polycentric accessibility to services than others. The disparity in the local availability of services is an additional layer of difference between cities, as some, despite an excellent average local accessibility, are highly heterogeneous.

Second, we asked ourselves about the possibility of improving present cities, considering the equality of access to urban opportunities. The first question has been whether a better geographical redistribution of the existing POIs would enhance the accessibility patterns for a larger fraction of the population. We ran this simulation through our original relocation algorithm for many different cities, and we concluded that relocating existing POIs (that is, without deploying new ones) would achieve the objective of substantially improving the proximity of peripheral and underserved areas while not impoverishing the central, already well-served, areas. This result advocates for more decentralization policies of the cities' activities, favoring populated peripheral areas.

Finally, we asked whether it is possible to conceive cities that would be optimal from the perspective of equal access to opportunities. In this case, we simulate scenarios of the geographical distribution of POIs by removing the constraints of the present distribution of POIs.

Instead, we simulated the best positioning of an increasing number of POIs starting from an empty city (that is, without POIs). This simulation allows us to identify, for each city, the minimum number of POIs needed for the city to be 15-minute for at least 90% of their residents. This observable nicely illustrates how hard it is for a city to become 15-minute, and how large the fluctuations are across cities. In some urban contexts, particularly in US cities, the number of POIs per capita would be too large to be economically sustainable. A scenario emerges in which the very notion of the 15-minute city can not be a one-fits-all solution and is not a viable option in areas with a too-low density and a pronounced sprawl. On the other side of the spectrum, in many dense cities, the number of POIs needed is relatively low and can thus be achieved with minimum effort from policy-makers. Paris, Barcelona, and Milan represent compelling examples demonstrating the efficacy of urban planning policies oriented towards local accessibility. On the other hand, there are major urban centers in the Global South, such as Nairobi, Mumbai and Addis Ababa, that currently have fewer POIs compared to the previous examples, but still demonstrate the potential for effective urban planning focused on local accessibility. Although our analysis on the current accessibility of African and developing cities can be biased by a lack of complete and reliable open data (see the Supplementary Section 4 for further discussion on this bias), the analysis of the optimal distribution reveals the potential of these dense cities to supply proximity services. These examples are potent arguments for the feasibility of 15-minute cities in dense urban areas and offer valuable lessons for other cities aiming to achieve similar outcomes. As for more sprawling cities, for instance, US cities, the low densities doom them far from the idea of a proximity-based city.

Since a substantial increase in population density in already inhabited areas is not an option in many cases, a more general theoretical framework[27] is appropriate to include local population densities as a relevant variable for non-compact urban contexts. In addition, distinguishing essential from non-essential services and their varying abundance is important to distinguish the feasibility of local proximity for categories of services. Only some of the POIs have the same degree of replication. One can easily open a new grocery store virtually everywhere, but one cannot replicate a landmark monument or major infrastructure (for example, hospital, opera theater, football stadium) in every neighborhood. For this reason, it is interesting to promote a slightly different notion of city in which what is optimized is not only the transit time but the amount of opportunities, wherever they are, available to residents.

Before concluding, let us remark that since our analysis is based on open data, it can be subject to some limitations depending on possible sources of bias. For instance, POIs data coverage can be limited in some areas or cities, or the walkability of some areas could be different in some cases[28]. Another limitation is that all categories of POIs are treated the same within and between different cities. Future work must consider climatic variations and cultural differences by adjusting the weights of different POI categories according to local preferences and needs. These minor issues can be progressively solved using more precise and complete data. The interested reader can find a more detailed discussion about possible critical points in Supplementary Section 4.

In conclusion, our research represents a crucial step forward in understanding urban accessibility and the viability of proximity-based cities. Although our findings might provide a new perspective on traditional planning approaches, they offer practical and actionable insights for designing sustainable, accessible and liveable urban environments. Our work also highlights that the ideal of a proximity-based city heralded by the 15-minute city must be balanced by faster and more reliable public transportation services that allow for eliminating some of the disparities in peripheries, particularly those due to the heterogeneity of local population densities. Finally, a broader perspective integrating socio-economic and cultural factors should be included as a next step to shift focus from only time-based to value-based cities. As we walk towards imagining future cities, we hope our research catalyzes deeper exploration and triggers concrete actions in urban planning and policy.

Methods

The data analysis was mainly performed using Python and the GeoPandas package[29], as well as Open Source Routing Machine (OSRM)[30] through calls to their API.

Data Acquisition

The data acquisition stage involved collecting four core datasets:

- City boundaries. The boundaries of the cities were acquired from Organisation for Economic Co-operation and Development (OECD) shapefiles[31], focusing on the defined core city. When OECD data were unavailable, we used the Global Human Settlement files[32], focusing on the city's core for the analysis.
- POIs. Amenities, which we generically refer to as POIs, were sourced from OpenStreetMap[33] and classified according to their tags. Only POIs representing services for people were retained, and non-service POIs such as trees and buildings were disregarded.
- Population data. Population data were derived from WorldPop34, providing demographic context for our accessibility measures. Our study used a 100 m population density grid corrected to match municipal UN population estimates35.
- Times of accessibility. The times of accessibility to POIs from points in the city were calculated using OSRM[30], based on OpenStreetMap data.

Data preparation

The data preparation stage encompassed two main steps:

- Hexagonal grid generation. A regular hexagonal grid with a side length of 200 m was generated across the city. Hexagons without any nearby POIs were omitted from further analysis.
- POI categorization. The retained POIs were categorized into nine service types: outdoor activities, learning, supplies, eating, moving, cultural activities, physical exercise, services and health care. This categorization allowed for both general and category-specific accessibility assessment.

Accessibility calculation

With the prepared data, we proceeded to compute the accessibility. Our measure of accessibility is similar to what has been proposed in the literature under the name 'dual access'[15]. We compute it as follows:

- Hexagon-level accessibility. For each hexagon, k, we identified the 20 nearest POIs in each category using OSRM for both walking and cycling routes. In the main text, we only show the results for the simulations for accessibility by walking, and we refer to the online platform for the simulations about accessibility by bike. The average time $\langle t\rangle_{c,k}$ required to access the 20 nearest POIs of category c in the hexagon k, is computed as follows:

$$\langle t\rangle_{c,k} = \frac{1}{n}\sum_{i=1}^{n} t_i^{c,k} \tag{1}$$

where n is the number of POIs (in this case, $n = 20$) and $t_i^{c,k}$ is the time required to reach the i-th POI of category c in the hexagon k. This procedure yields the accessibility value for that category in that hexagon, and it is equivalent to calculating the dual access to the 20 closest POIs for each hexagon. We selected a threshold of 20 POIs to account for choice in accessing services. A small amount of theoretically accessible places can be seen as too restrictive and enhance negative feelings towards proximity-based cities in the general public. In Supplementary Section 6, we show that changing the number of accessible POIs does not alter the main results of our work, and the PT varies homogeneously in cities.

- City-level accessibility. The accessibility measures for each hexagon k were averaged to generate a single proximity time index PT_k for that hexagon:

$$\mathrm{PT}_k = \frac{1}{m}\sum_{c=1}^{m}\langle t\rangle_{c,k} \tag{2}$$

where m is the number of categories. These indices were then averaged across the city, weighted by the population p_k within each hexagon, to compute the overall city accessibility $\mathrm{PT}_{\mathrm{city}}$:

$$\mathrm{PT}_{\mathrm{city}} = \frac{\sum_{k=1}^{K}\mathrm{PT}_k p_k}{\sum_{k=1}^{K} p_k} \tag{3}$$

where K is the number of hexagons, PT_k is the proximity time index of the k-th hexagon, and p_k is the population of the k-th hexagon. The corresponding category measures were averaged to derive specific city scores for individual categories.

- 15-minute accessibility. We also computed the percentage of the city's population within a 15-minute access range, providing another measure of city-wide accessibility:

$$F_{15} = \frac{\sum_{k=1,\mathrm{PT}_k\le 15}^{K} p_k}{\sum_{k=1}^{K} p_k}\times 100 \tag{4}$$

Inequality assessment

Lastly, to measure the level of inequality in accessibility across the city, we calculated the Gini inequality index G:

$$G = 1 - \frac{2\sum_{p=1}^{N_{\mathrm{pop}}}\sum_{p'\le p}\mathrm{PT}_{p'}}{N_{\mathrm{pop}}\sum_{p=1}^{N_{\mathrm{pop}}}\mathrm{PT}_p} \tag{5}$$

where PT_p is the p-th proximity time measure of a single person in the city when sorted in non-decreasing order and N_{pop} is the total population of a city.

This metric provides a non-trivial measure of urban accessibility, revealing how unequal local accessibility is in the city. Anyway, a compact city with a good proximity time will not necessarily have a low Gini coefficient because the accessibility times can still fluctuate.

POI relocation algorithm

Our algorithm takes the population distribution within the city's hexagonal grid as input and optimizes the allocation of POIs independently of the category. The optimization is performed separately for each category. Let us briefly describe the rationale of the algorithm, whose steps are depicted in Fig. 2. The optimization aims to keep the number of POIs per capita constant in the city to serve every area equally, considering its population density. Therefore, each POI will have to serve on average the same number of people, which we call capacity CAP and is a global quantity of the considered city and category:

$$\mathrm{CAP}_c = \frac{N_{\mathrm{pop}}}{N_c}, \tag{6}$$

where c is a category of services and N_c is the number of POIs of category c in the city.

The inverse of the capacity is the number of POIs per capita, reading:

$$P_c = \frac{N_c}{N_{\mathrm{pop}}}. \tag{7}$$

Note that we use the number of POIs per thousand people instead of POIs per capita to improve the measure's readability and meaning.

A city is initially considered empty and then iteratively filled with POIs. Each iteration places a POI in the hexagon with the highest population demand (that is, population needing a POI) that can be reached within a chosen time threshold, 15 minutes in our case. We make the algorithm slightly random by randomly choosing a neighbor of the highest demand hexagon to avoid the effect of always choosing the cells with the locally highest population in need. After placing the POI, a number of people equal to the POI capacity CAP_c will be deducted from the people in need in the hexagons that can reach the POI in 15 minutes, in amounts proportional to the population demands of the hexagons. At the end of this iteration, the total sum of the needing population is reduced by exactly CAP_c. This operation is iterated until all POIs have been assigned to a hexagon, yielding an equal distribution of services per capita across the city. The algorithm is the same for all categories, and it is dependent only on the city's population distribution. The number of POIs per capita in each 15-minute neighborhood should then be (roughly) the same across the city and equal to the average POIs per capita in the city (Fig. 2). A pseudocode for the algorithm can be found in Supplementary Section 5.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in the analysis is open source and referenced in the Methods. The data supporting the findings of this study are available from the corresponding author upon request.

Code availability

The code used in the analysis is described as pseudo-code in the Supplementary Information and can be made available upon reasonable request.

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Acknowledgements

The authors thank B. Monechi and C. Chiappetta for their contributions.

Author contributions

All authors designed the research and analysis. B.C. and M.B. performed the data analysis. V.L., M.B. and H.P.M.M. wrote the paper. All authors revised the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44284-024-00119-4.

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Peer review information *Nature Cities* thanks Elsa Arcaute, Gerhard Schmitt and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Last updated by author(s): Jul 25, 2024

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Data collection	Data collected from open sources through basic Python codes or manually downloaded. The data sources are accurately referenced in the text.
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Study description	The study analyses accessibility in many worldwide cities and proposes a framework to implement equal accessibility to proximity services. By leveraging this methodology the study proposes a metric for a potential to become a 15-minute city.
Research sample	Cities from all over the world were analysed. For the optimisation strategy and analysis of the potential to be 15-minute, about 50 cities were selected from all continents and with different characteristics. The authors limited the study to cities that have clearly defined resonable city boundaries, as the choice of the city limits could alter the result of a city (for instance selecting a functional urban area or its urban core).
Sampling strategy	The cities examined were chosen based on interest, relative reliability of the data, and diversity. The cities are selected mixing different cultures and urbanist styles, date of foundation and of renovation, different socioeconomic statuses. When performing simulations on the number of POIs to put in a city, from a small number the simulations were stopped when almost all the population would have a reasonable accessibility, and in particular when the threshold of 90% of the population was reached. The simulations have very little randomness so repeating them 10 times was sufficient to smooth out the results.
Data collection	All data used is based on open source datasets clearly cited in the main text, downloaded manually.
Timing	Data was collected from the referenced online sources in the first 6 months of 2023, between Jan 1st and June 30th.
Data exclusions	No data were excluded
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